

Chapter 1: Introduction

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Introduction

What Operating Systems Do:

- Computer-System Organization
- Computer-System Architecture
- Operating-System Structure
- Operating-System Operations
- Process Management
- Memory Management
- Storage Management
- Protection and Security
- Kernel Data Structures
- Computing Environments





Objectives

- To describe the basic organization of computer systems
- To provide a grand tour of the major components of operating systems
- To give an overview of the many types of computing environments





What is an Operating System?

- A program that acts as an **intermediary** between a user of a computer and the computer hardware.
- Operating system goals:
 - Execute user programs and make solving user problems **easier**.
 - Make the computer system **convenient** to use.
- Use the computer hardware in an **efficient** manner.





Software

The two most common types of software are :

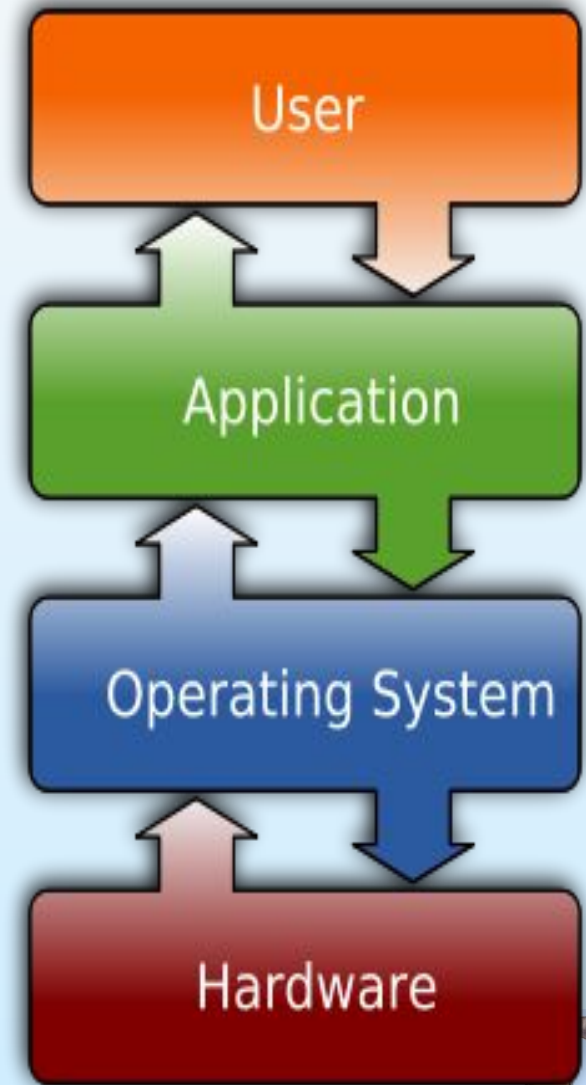
- System software
- Application software.

What is System Software?

System Software refers to the operating system and all utility programs that manage computer resources at a low level. Systems software includes compilers, loaders, linkers, and debuggers.

What is Application Software?

Applications software comprises programs designed for an end user, such as word processors, database systems, and spreadsheet programs.





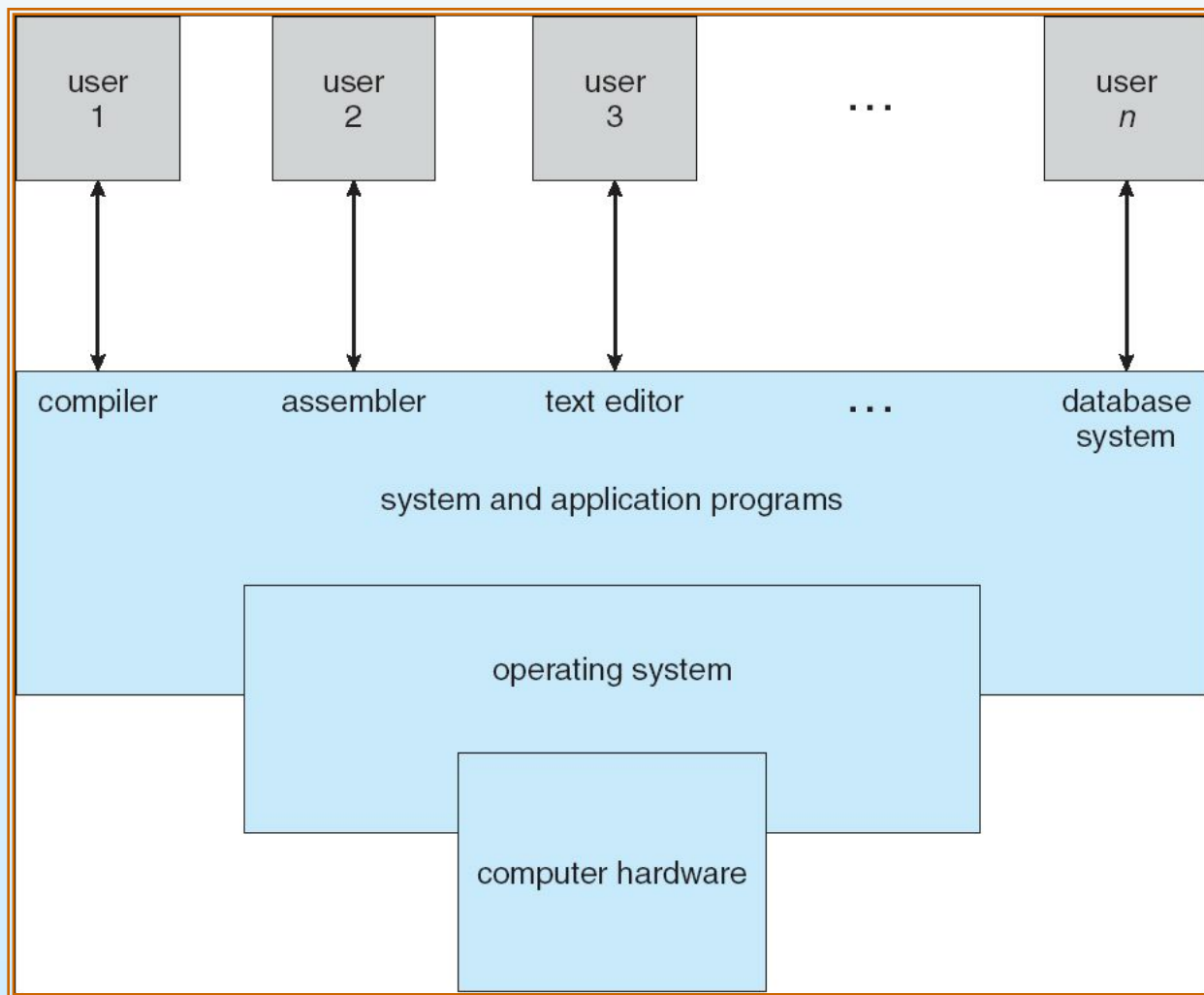
Computer System Structure

- Computer system can be divided into four components
 - **Hardware** – provides basic computing resources
 - 4 CPU, memory, I/O devices
 - **Operating system**
 - 4 Controls and coordinates use of hardware among various applications and users
 - **Application programs** – define the ways in which the system resources are used to solve the computing problems of the users
 - 4 Word processors, compilers, web browsers, database systems, video games
 - **Users**
 - 4 People, machines, other computers





Four Components of a Computer System





Operating System Definition

- OS is a **resource allocator**
 - Manages all resources
 - Decides between conflicting requests for efficient and fair resource use
- OS is a **control program**
 - Controls execution of programs to prevent errors and improper use of the computer





Operating System Definition (Cont.)

- No universally accepted definition
- “Everything a vendor ships when you order an operating system” is good approximation
 - But varies wildly
- “The one program running at all times on the computer” is the **kernel**. Everything else is either a system program (ships with the operating system) or an application program





Computer Startup

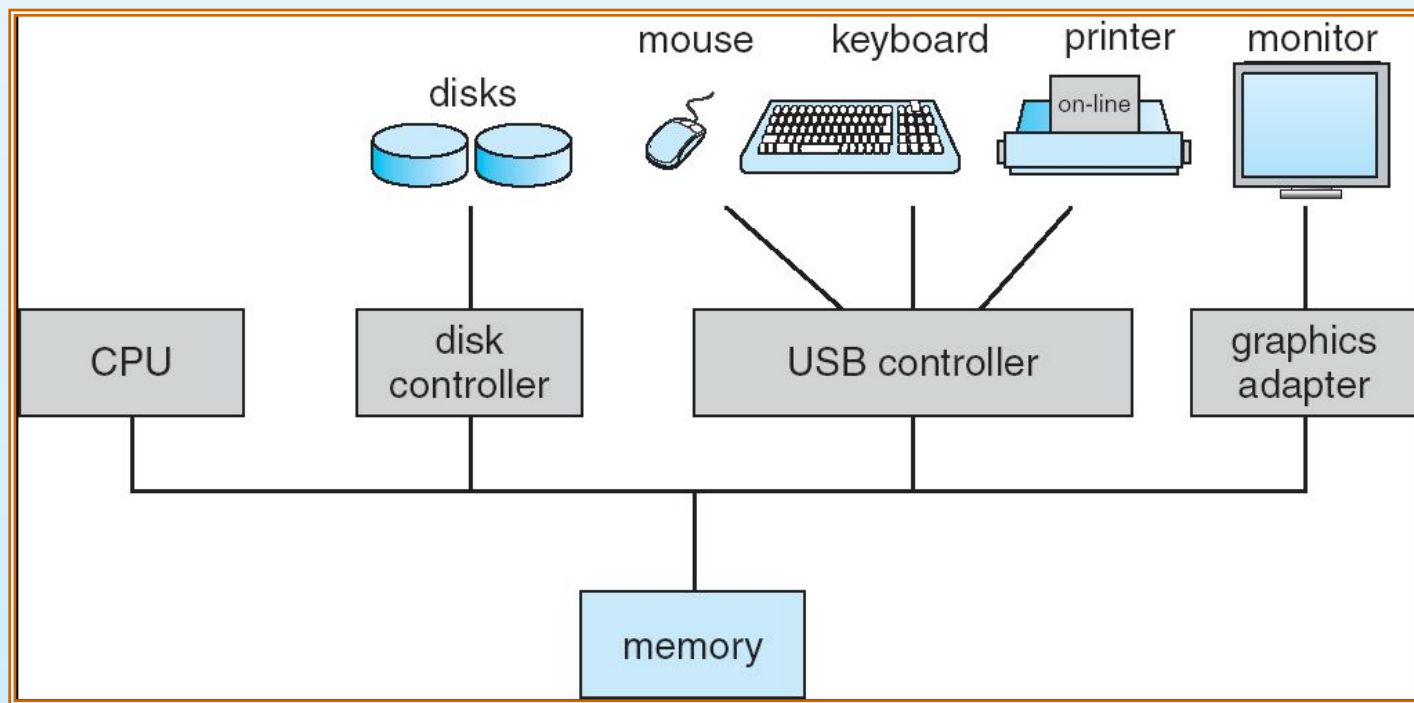
- **bootstrap program** is loaded at power-up or reboot
 - Typically stored in ROM or EPROM, generally known as **firmware**
 - Initializes all aspects of system
 - Loads operating system kernel and starts execution





Computer System Organization

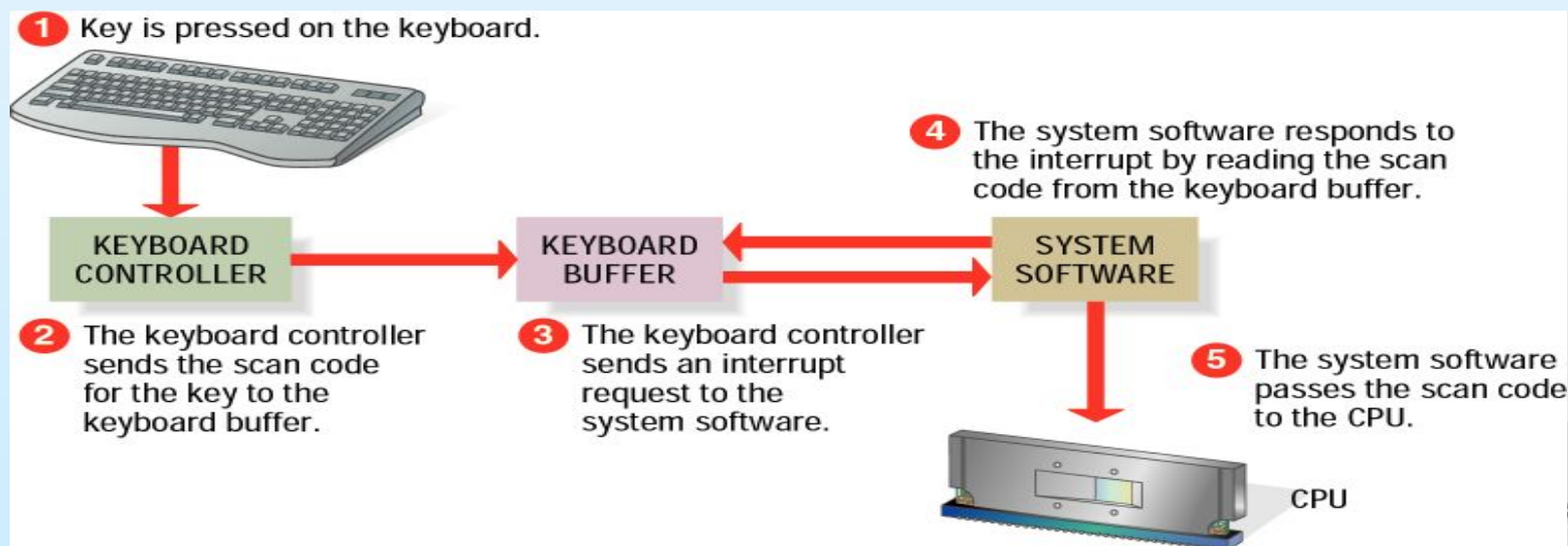
- Computer-system operation
 - One or more CPUs, Memory, device controllers connect through a common bus which is called system bus.





Computer-System Operation

- I/O devices and the CPU can execute **concurrently**.
- Each **device controller** is in charge of a particular device type.
- Each device controller has a local buffer.
- CPU moves data from/to main memory to/from local buffers
- Device controller informs CPU that it has finished its operation by causing an *interrupt*.





Common Functions of Interrupts

- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector**, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A **trap** or **exception** is a software-generated interrupt caused either by an error or a user request
- An operating system is **interrupt driven**





Interrupt Handling

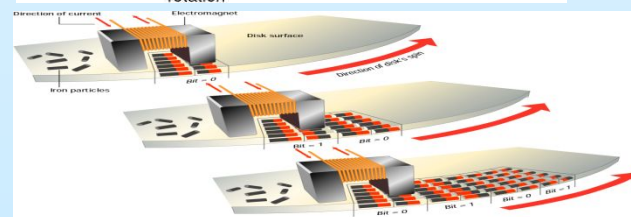
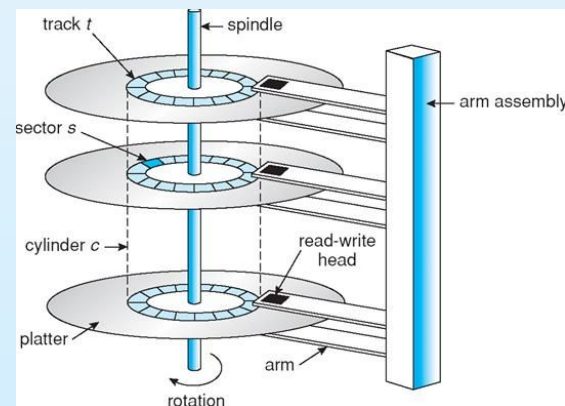
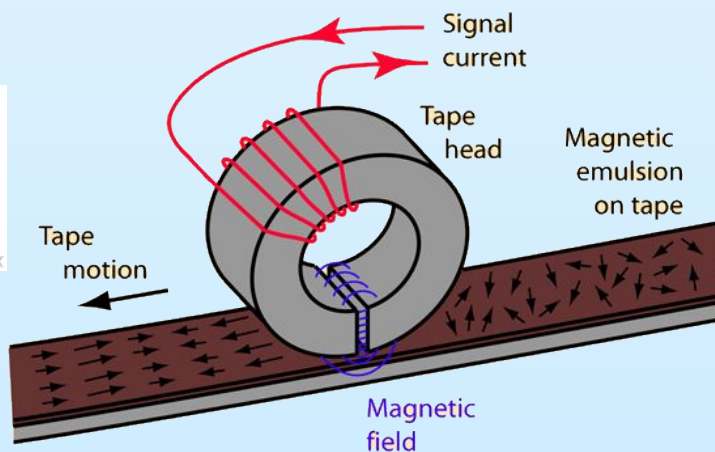
- The operating system preserves the state of the CPU by storing registers and the program counter
- Determines which type of interrupt has occurred:
 - **polling**
 - **vectored** interrupt system
- Separate segments of code determine what action should be taken for each type of interrupt





Storage Structure

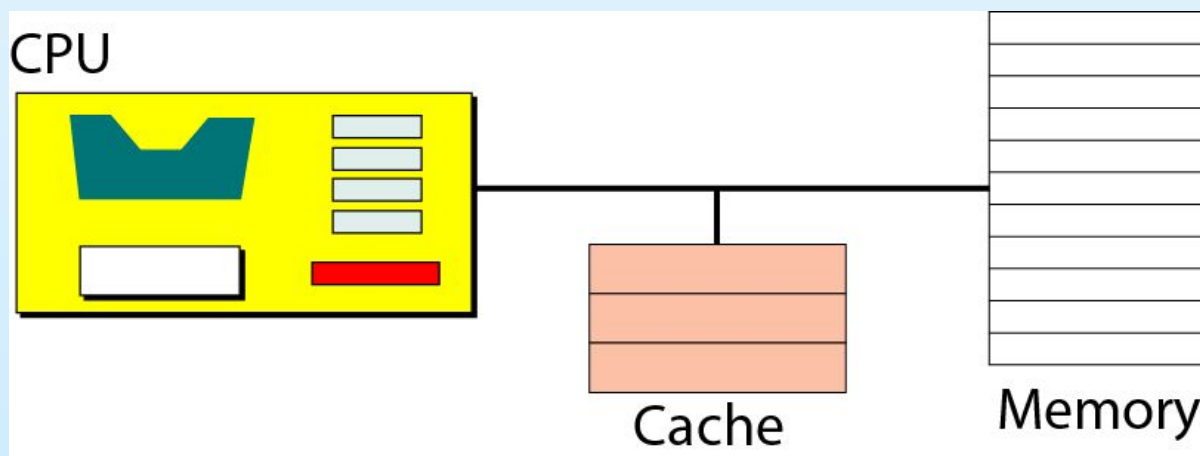
- **Main memory** – only large storage media that the CPU can access directly.
- **Secondary storage** – extension of main memory that provides large nonvolatile storage capacity.
 - **Magnetic disks** – rigid metal or glass platters covered with magnetic recording material
 - 4 Disk surface is logically divided into *tracks*, which are subdivided into *sectors*.
 - 4 The *disk controller* determines the logical interaction between the device and the computer.
 - **Magnetic Tapes**





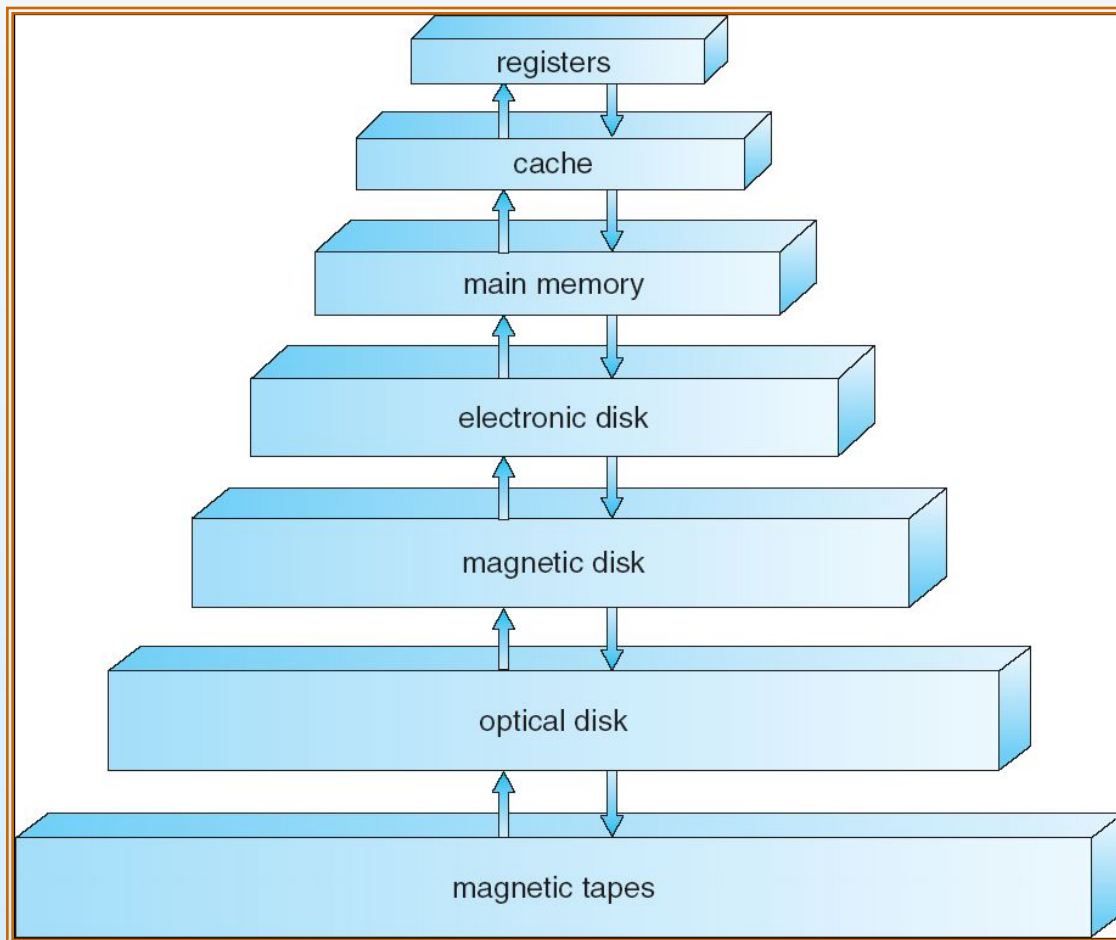
Storage Hierarchy

- Storage systems organized in **hierarchy**.
 - Speed
 - Cost
 - Volatility
 - Size
- *Caching* – Information in use copied from slower to faster storage temporarily





Storage-Device Hierarchy





Performance of Various Levels of Storage

- Movement between levels of storage hierarchy can be explicit or implicit

Level	1	2	3	4
Name	registers	cache	main memory	disk storage
Typical size	< 1 KB	> 16 MB	> 16 GB	> 100 GB
Implementation technology	custom memory with multiple ports, CMOS	on-chip or off-chip CMOS SRAM	CMOS DRAM	magnetic disk
Access time (ns)	0.25 – 0.5	0.5 – 25	80 – 250	5,000.000
Bandwidth (MB/sec)	20,000 – 100,000	5000 – 10,000	1000 – 5000	20 – 150
Managed by	compiler	hardware	operating system	operating system
Backed by	cache	main memory	disk	CD or tape





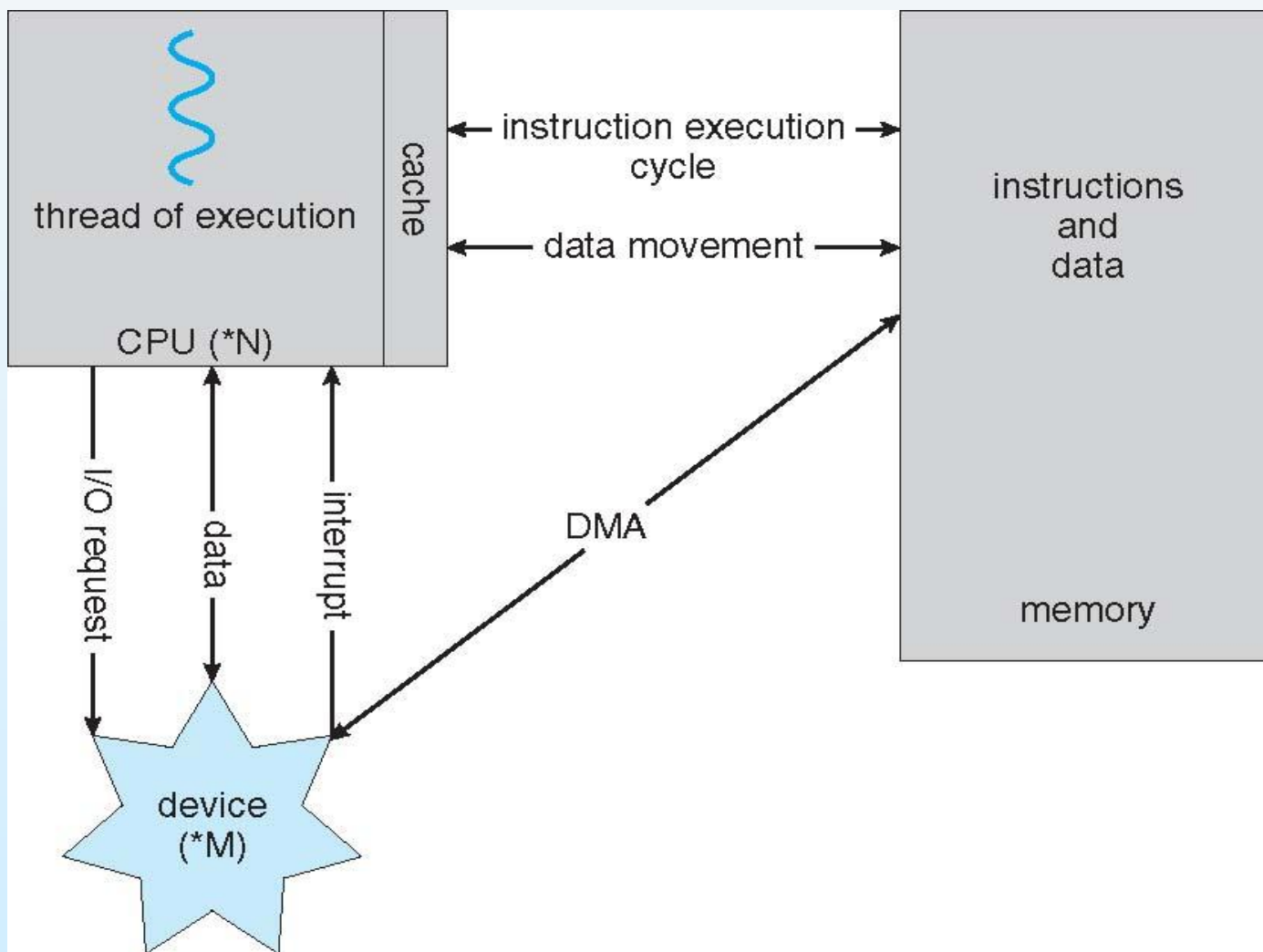
Direct Memory Access Structure

- Used for high-speed I/O devices able to transmit information at close to memory speeds
- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention
- Only one interrupt is generated per block, rather than the one interrupt per byte





How a Modern Computer Works





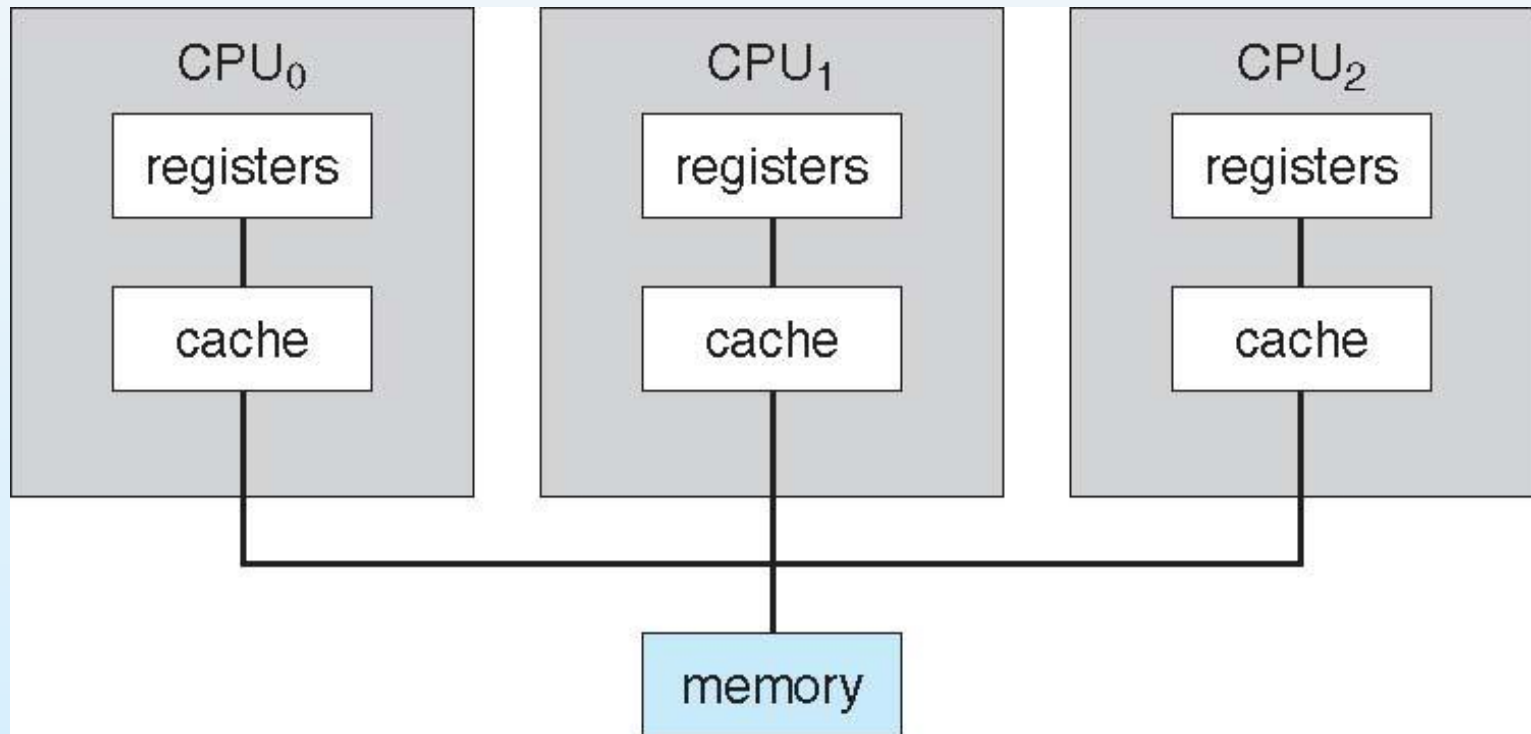
Computer-System Architecture

- Most systems use a single general-purpose processor
 - Most systems have special-purpose processors as well
- **Multiprocessors** systems growing in use and importance
 - Also known as **parallel systems**, **tightly-coupled systems**
 - Advantages include:
 1. **Increased throughput**
 2. **Economy of scale**
 3. **Increased reliability** – graceful degradation or fault tolerance
 - Two types:
 1. **Asymmetric Multiprocessing** – each processor is assigned a specific task.
 2. **Symmetric Multiprocessing** – each processor performs all tasks





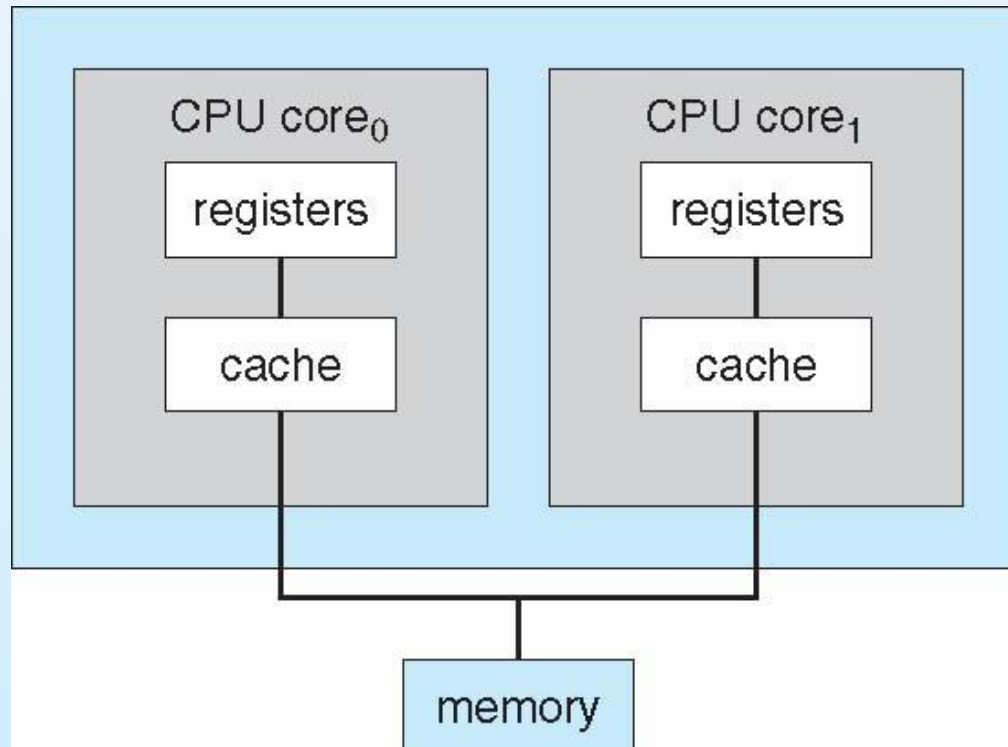
Symmetric Multiprocessing Architecture





A Dual-Core Design

- Multi-chip and **multicore**
- Systems containing all chips
 - Chassis containing multiple separate systems





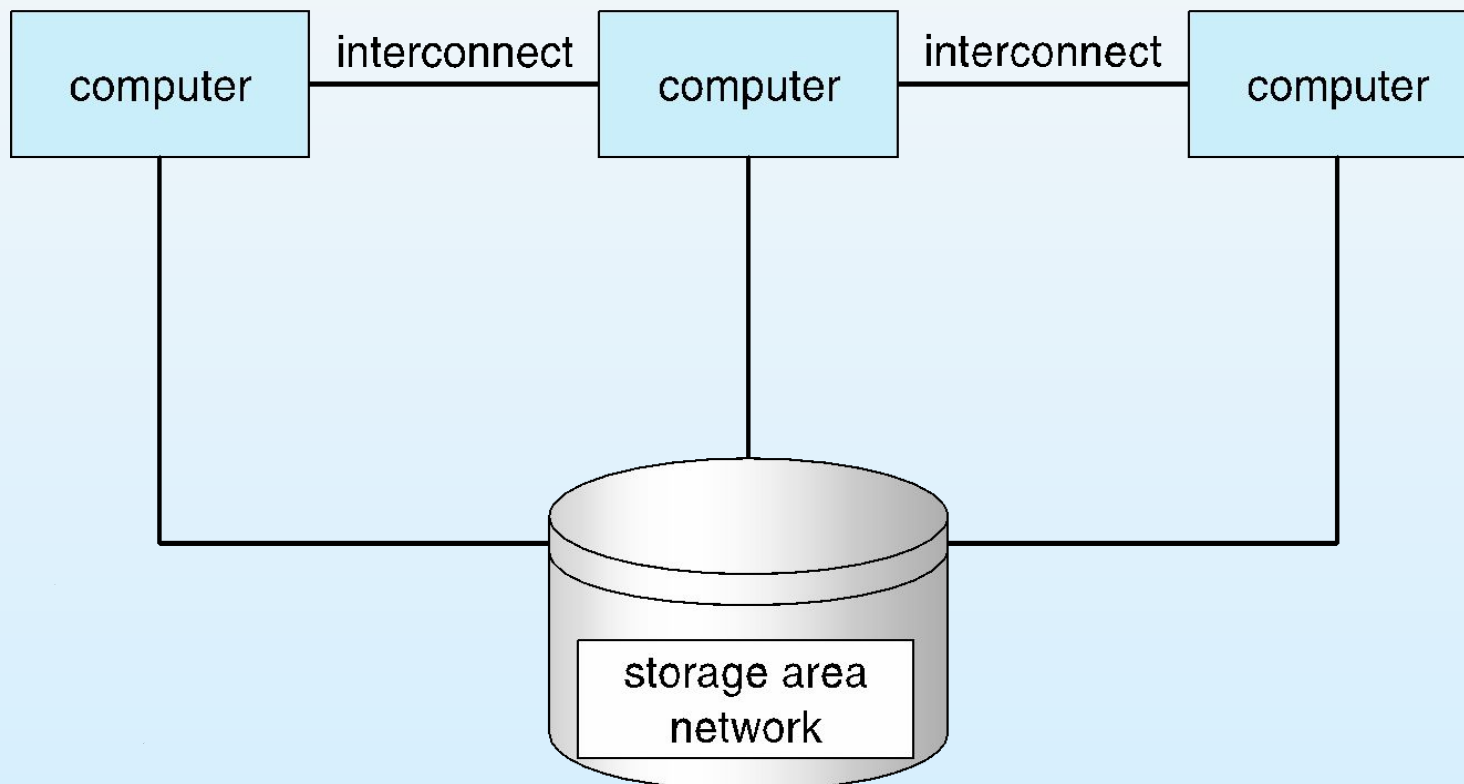
Clustered Systems

- Like multiprocessor systems, but multiple systems working together
 - Usually sharing storage via a **storage-area network (SAN)**
 - Provides a **high-availability** service which survives failures
 - 4 **Asymmetric clustering** has one machine in hot-standby mode
 - 4 **Symmetric clustering** has multiple nodes running applications, monitoring each other
 - Some clusters are for **high-performance computing (HPC)**
 - 4 Applications must be written to use **parallelization**
 - Some have **distributed lock manager (DLM)** to avoid conflicting operations





Clustered Systems





Operating System Structure

- **Multiprogramming** needed for efficiency
 - Single user cannot keep CPU and I/O devices busy at all times
 - Multiprogramming organizes jobs (code and data) so CPU always has one to execute
 - A subset of total jobs in system is kept in memory
 - One job selected and run via **job scheduling**
 - When it has to wait (for I/O for example), OS switches to another job





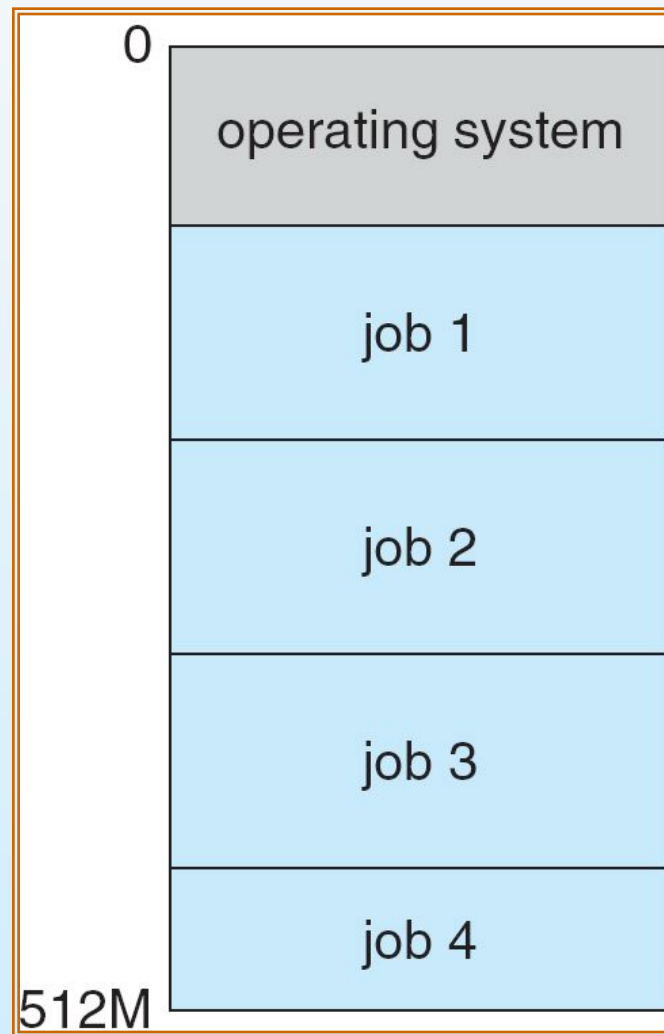
Operating System Structure (cont)

- **Timesharing (multitasking)** is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive** computing
 - **Response time** should be < 1 second
 - Each user has at least one program executing in memory
 - **process**
 - If several jobs ready to run at the same time □ **CPU scheduling**
 - If processes don't fit in memory, **swapping** moves them in and out to run
 - **Virtual memory** allows execution of processes not completely in memory





Memory Layout for Multiprogrammed System





Operating-System Operations

- With sharing many processes could be adversely affected by a **bug** in one program. So, a properly designed OS must ensure that an **incorrect program** can not run and also can not cause other programs to execute incorrectly.
- Many errors detected by hardware can be handled by OS.
- Hardware generates **interrupt**.
- Software error handled by **exception** or **trap**.





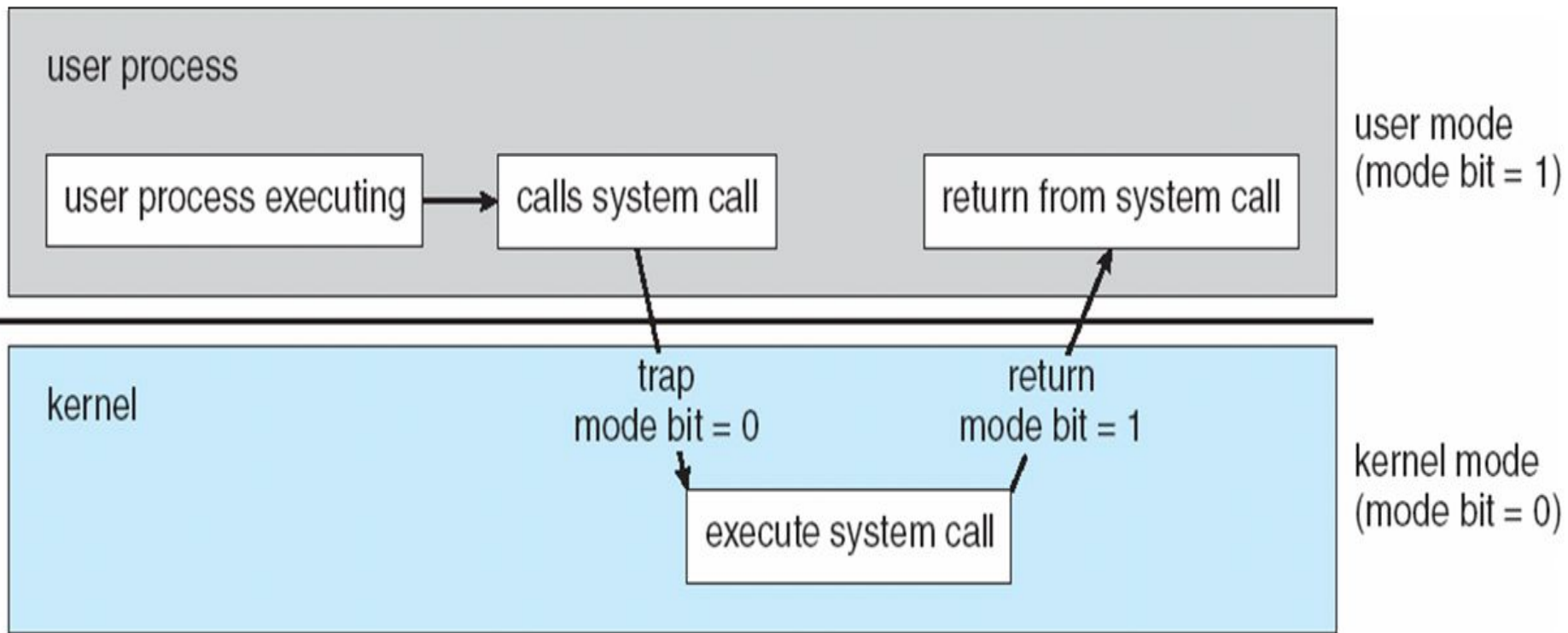
Operating-System Operations

- **Dual-mode** operation allows OS to protect itself and other system components
 - **User mode (1)** and **Monitor/System mode/ Kernel mode (0)**
 - **Mode bit** provided by hardware
 - 4 Provides ability to distinguish when system is running user code or system code
 - 4 Some instructions designated as **privileged**, only executable in system mode





Transition from User to Kernel Mode





Process Management

The operating system is responsible for the following activities in connection with process management:

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling





Memory Management

- To execute a program all (or part) of the instructions must be in memory
- All (or part) of the data that is needed by the program must be in memory.
- Memory management determines what is in memory and when
 - Optimizing CPU utilization and computer response to users
- Memory management activities
 - Keeping track of which parts of memory are currently being used and by whom
 - Deciding which processes (or parts thereof) and data to move into and out of memory
 - Allocating and de-allocating memory space as needed





Storage Management

- File-System management
 - Files usually organized into directories
 - Access control on most systems to determine who can access what
 - OS activities include
 - 4 Creating and deleting files and directories
 - 4 Primitives to manipulate files and directories
 - 4 Mapping files onto secondary storage
 - 4 Backup files onto stable (non-volatile) storage media





Protection and Security

- **Protection** – any mechanism for controlling access of processes or users to resources defined by the OS
- **Security** – defense of the system against internal and external attacks
 - Huge range, including denial-of-service, worms, viruses, identity theft, theft of service
- Systems generally first distinguish among users, to determine who can do what
 - User identities (**user IDs**, security IDs) include name and associated number, one per user
 - User ID then associated with all files, processes of that user to determine access control
 - Group identifier (**group ID**) allows set of users to be defined and controls managed, then also associated with each process, file
 - **Privilege escalation** allows user to change to effective ID with more rights

